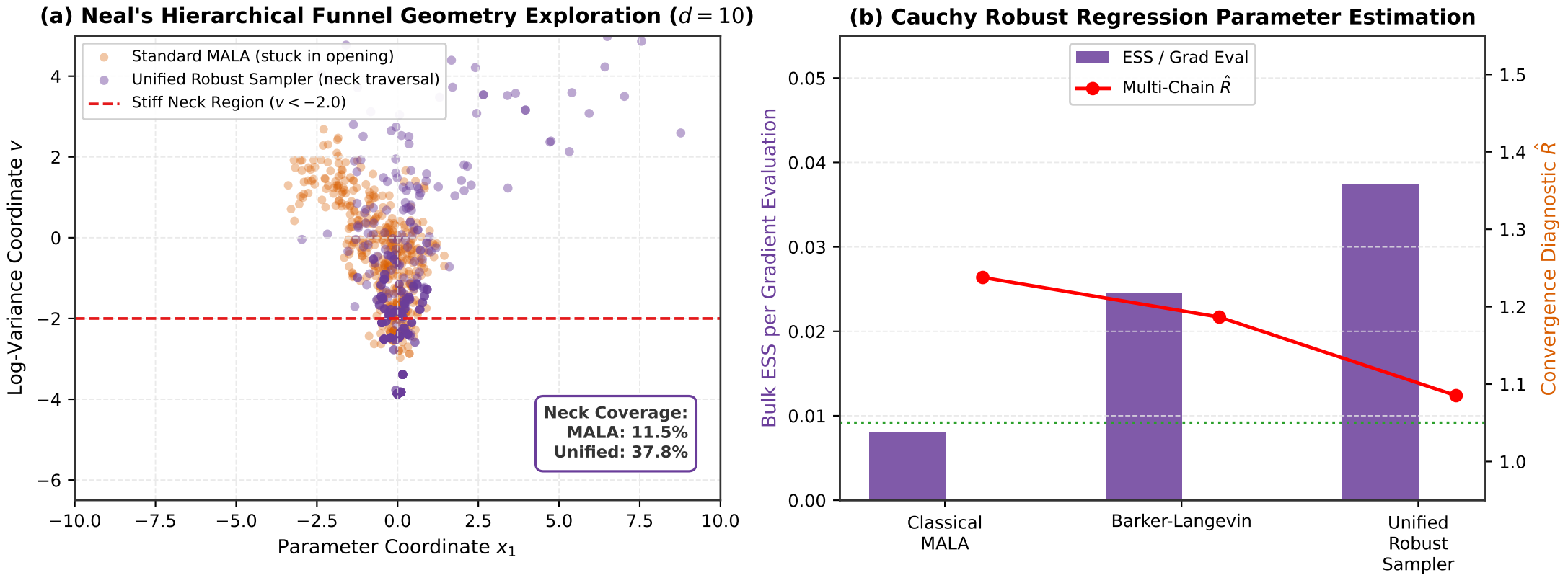
(a) Neal's Hierarchical Funnel Geometry Exploration ($d = 10$)
Standard MALA (stuck in opening)
Unified Robust Sampler (neck traversal)
Stiff Neck Region ($v < -2.0$)
Neck Coverage:
MALA: 11.5%
Unified: 37.8%
Log-Variance Coordinate v
Parameter Coordinate x_1
4
2
0
−2
−4
−6
−10.0
−7.5
−5.0
−2.5
0.0
2.5
5.0
7.5
10.0
(b) Cauchy Robust Regression Parameter Estimation
ESS / Grad Eval
Multi-Chain $\hat{R}$
Bulk ESS per Gradient Evaluation
Convergence Diagnostic $\hat{R}$
0.05
0.04
0.03
0.02
0.01
0.00
1.5
1.4
1.3
1.2
1.1
1.0
Classical MALA
Barker-Langevin
Unified Robust Sampler